Relational Quantum Memory and Operational Access

A consciousness-motivated model with an exact resource-bounded benchmark

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Abstract

We specify a finite-dimensional quantum instrument model coupling an active system to memory and classical feedback. Its motivation is the distinction between a presently available report and information that can influence later reports after an intervention. Elementary propositions establish hidden-state discrimination, its degradation under dephasing, and the accessibility of correlations invisible in subsystem marginals. We then resolve a concrete resource-bounded comparison: a two-bit random access task implemented with one classical bit or one qubit. Complete proofs give the optimal scores $3/4$ and $(1 + 1/\sqrt{2})/2$. For an isotropically noisy quantum protocol we compute the exact worst-input total-variation distance to the shared-randomness one-bit classical class, $\max\{0, \eta/(2\sqrt{2}) - 1/4\}$. A finite-tree classical simulation theorem explains why this separation requires an enforced communication or memory bottleneck. The benchmark is established random-access-code theory, not a claimed new discovery. The proposed link between operational accessibility and conscious processing is a separate empirical hypothesis. No neural data, physical identification of a quantum substrate, or formal Lean verification of this manuscript is supplied.

1 Scope and contribution status

Quantum instrument models of cognition and quantum processes with memory already have substantial mathematical frameworks; see [3, 2]. This note supplies a self-contained synthesis and a fully specified benchmark rather than a claim of a new fundamental theory. The random-access-code separation is known [1]; all benchmark proofs are included so that the assumptions and conclusions can be audited directly. The noisy distance formula below is presented as a corollary of that benchmark, with no priority claim.

Three assertions must be distinguished: existence of a mathematical model; realization of its dynamics in a physical device; and identification of those dynamics with conscious experience. The first can be established here. The latter two require independent evidence. In particular, the model contains no theorem about the existence of God, the generation of spacetime, or a clinical explanation of psychosis. Neither philosophical motivation nor a formal proof of an information-theoretic statement supplies these additional identifications.

2 Minimal dynamical model

Let $\mathcal{H}_S$ and $\mathcal{H}_M$ be finite-dimensional Hilbert spaces for active processing and memory. A state is $\rho \geq 0$ on $\mathcal{H}_S \otimes \mathcal{H}_M$, with $\operatorname{tr} \rho = 1$. A question or intervention a selects an instrument $\{\mathcal{I}_y^a\}_y$: each map is completely positive and trace-nonincreasing, and their sum is trace-preserving. The elementary cycle is

$$p(y \mid \rho, a) = \operatorname{tr} \mathcal{I}_y^a(\rho), \qquad \rho' = \frac{\mathcal{I}_y^a(\rho)}{p(y \mid \rho, a)}. \tag{1}$$

The conditional state is used only when the denominator is positive. A classical policy chooses the next intervention from the recorded history $h = (a_0, y_0, \ldots, a_{t-1}, y_{t-1})$. This is ordinary causal feedback, not backwards-in-time influence.

A two-qubit realization uses Pauli matrices X, Y, Z and

$$H_a = \frac{\hbar}{2}(\omega_S Z \otimes I + \omega_M I \otimes Z + gX \otimes X + u_a Y \otimes I), \tag{2}$$

$$U_a = e^{-i\Delta t H_a/\hbar}, \tag{3}$$

$$\mathcal{D}_\lambda(\rho) = (1-\lambda)\rho + \lambda \sum_{j=0}^{1} (P_j \otimes I)\rho(P_j \otimes I), \quad P_j = |j\rangle\langle j|, \tag{4}$$

$$\mathcal{I}_y^a(\rho) = (K_y^a \otimes I)\mathcal{D}_\lambda(U_a \rho U_a^\dagger)(K_y^{a\dagger} \otimes I), \quad \sum_y K_y^{a\dagger} K_y^a = I. \tag{5}$$

Here $0 \le \lambda \le 1$, and the coefficients in $H_a/\hbar$ have units of inverse time. These formulas define valid instruments: unitary conjugation and dephasing are channels, and the last normalization condition makes the summed measurement map trace-preserving. They are a parameterized architecture, not a fit to biological data.

3 Information unavailable to one question

Proposition 3.1. *Let $r > 0$, $r^2 + z^2 \le 1$, and $\rho_\pm = (I + zZ \pm rX)/2$. Measurement in the Z basis gives identical distributions for the two states. After the Hadamard unitary W, the probability of outcome zero differs by r.*

Proof. The eigenvalues of each state are $(1 \pm \sqrt{r^2 + z^2})/2$, hence both are valid. Initially $\mathrm{tr}(P_0\rho_\pm) = (1+z)/2$. Since $WXW^\dagger = Z$ and $WZW^\dagger = X$, we have

$$W\rho_\pm W^\dagger = (I + zX \pm rZ)/2, \qquad \mathrm{tr}(P_0 W \rho_\pm W^\dagger) = (1 \pm r)/2.$$

□

The comparison is between alternative protocols on identically prepared ensembles. An actual preceding sharp Z measurement removes the relevant coherence and changes the experiment.

Proposition 3.2 (Dephasing and optimal discrimination)**.** *After Z-dephasing with strength λ, the optimal probability of distinguishing ρ_+ from ρ_- with equal priors is*

$$P_{\mathrm{succ}} = \frac{1 + (1-\lambda)r}{2}.$$

Proof. The state difference becomes $\Delta = (1-\lambda)rX$. A binary measurement with effect $0 \le E \le I$ has success probability $[1 + \mathrm{tr}(E\Delta)]/2$. The eigenvalues of Δ are $\pm(1-\lambda)r$; its positive eigenspace maximizes the trace and attains the stated value. □

Proposition 3.3 (Information in relations)**.** *There are orthogonal joint states with identical marginals on both subsystems that a joint operation converts into distinguishable local records.*

Proof. Take $|\Phi_\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$. Each marginal is $I/2$. The unitary $V = (W \otimes I)\,\mathrm{CNOT}$ sends these states to $|00\rangle$ and $|10\rangle$, respectively. Measurement of the first qubit distinguishes them perfectly. □

These are familiar quantum-information effects. They establish what the model can do, not that a brain implements the model.

4 Operational accessibility and the consciousness hypothesis

Definition 4.1. For a specified family of allowed finite protocols Π, define

$$\mathcal{A}_\Pi(\rho,\sigma) = \sup_{\pi\in\Pi} \|P_\rho^\pi - P_\sigma^\pi\|_{\mathrm{TV}}, \qquad \|p-q\|_{\mathrm{TV}} = \frac{1}{2}\sum_y |p(y)-q(y)|.$$

Here P_ρ^π is the distribution of the full classical record under protocol π.

Proposition 3.1 gives $\mathcal{A}_{\{Z\}} = 0$ and $\mathcal{A}_{\{W\to Z\}} = r$ for its pair of states. The quantity depends on the preparations and allowed access, not solely on an intrinsic property of one state.

Hypothesis 4.2 (Relational accessibility)**.** Some experimentally identified features of conscious processing may be predicted by the ability to convert distinctions stored in system–memory relations into records usable for subsequent control.

This is deliberately weaker than identifying consciousness with a density matrix. To test it, one must independently specify behavioral or physiological endpoints, a mapping from interventions to instruments, parameter-estimation procedures, and competing models. Prediction of those endpoints on held-out data is required. The scalar $\mathcal{A}_\Pi$ is not by definition a consciousness measure.

5 Classical records and the Shannon interface

For a history h and action a, the instrument model induces a classical kernel $K(y \mid h,a)$. Shannon entropy measures uncertainty in records, e.g.

$$H(Y_t \mid h,a) = -\sum_y K(y \mid h,a)\log_2 K(y \mid h,a).$$

A deterministic answer has zero entropy even if internal states remain different. Thus zero report entropy does not imply zero memory, zero coherence, or zero subjective value.

Theorem 5.1 (Finite-tree universal classical realization)**.** *Fix a finite tree of full histories with finite action and outcome sets and a kernel at every history–action pair. There is a single finite classical random seed and deterministic causal decoder reproducing the prescribed record law under every deterministic adaptive policy, with the same decoder at shared histories.*

Proof. Independently sample $S_{h,a} \sim K(\cdot \mid h,a)$ for every history–action pair, including arbitrary specified kernels on otherwise unreachable branches. Let S be the resulting finite table. At (h,a) output $S_{h,a}$. Reaching this pair depends only on earlier table entries, so its entry retains its specified conditional law. Multiplication of these conditional laws yields the prescribed probability of every transcript. The table and lookup rule do not depend on the policy. □

This construction assumes a global decoder that can consult the full history and an unrestricted table. It supplies existence, not optimal seed entropy. Optimizing $H(S)$ under simultaneous causal compatibility is a separate causal-seed problem; no sharp entropy theorem from an external Lean repository is assumed in the proofs here.

For distributions P, Q on finite transcripts, assume $Q(x) > 0$ whenever $P(x) > 0$. Write $Q(x_{1:n}) = \prod_t Q_t(x_t \mid x_{<t})$. Then

$$R_n = \log_2 \frac{P(X_{1:n})}{Q(X_{1:n})}, \tag{6}$$

$$\mathbb{E}_P R_n = D_{\mathrm{KL},2}(P\|Q) = \sum_{t=1}^{n} \mathbb{E}_P D_{\mathrm{KL},2}(P(\cdot \mid X_{<t})\|Q_t(\cdot \mid X_{<t})). \tag{7}$$

The identity follows by factoring both joint laws, taking logarithms, and conditioning each summand on the past. Individual realized R_n can be negative; its expectation is nonnegative. It is expected ideal code-length excess in bits, not automatically thermodynamic energy or the communication cost of repairing a simulator. Such a repair claim needs an additional operational theorem.

6 A complete resource-bounded separation

6.1 Task and resource contract

An encoder receives $x = (x_1, x_2)$ uniformly from $\{0,1\}^2$. After encoding, an independent query $j \in \{1,2\}$ is given to a decoder. The decoder outputs b and succeeds when $b = x_j$. Its mean score is

$$s(P) = \frac{1}{8} \sum_{x,j} P(b = x_j \mid x, j).$$

The classical class $\mathcal{C}_1$ permits one transmitted bit, unlimited computation, and arbitrary shared randomness independent of x, j. The query is unavailable to the encoder. The decoder has no input-dependent side channel or retained copy of x. Private randomness is allowed. The quantum class replaces the one-bit message by one qubit and permits arbitrary binary decoding POVMs. No preshared entanglement is allowed. Shared classical randomness is allowed.

Equivalently, this is a temporal memory task if the message is the only input-dependent state retained across the encoding/query boundary. In that interpretation the one-bit restriction applies to *all* input-dependent retained information. A two-qubit system is not automatically subject to this one-qubit benchmark; it must be restricted or reset so that only the permitted carrier crosses the boundary.

Theorem 6.1 (Optimal classical score)**.** *Every $Q \in \mathcal{C}_1$ satisfies $s(Q) \le 3/4$, and equality is attainable.*

Proof. Condition on all random coins, which are independent of the inputs. The strategy is then deterministic. Each message $m \in \{0,1\}$ determines a predicted pair $c_m = (g(m,1), g(m,2))$. For input x, at least $\min_m d_H(x, c_m)$ coordinates are wrong even with the optimal encoding. With at most two distinct codewords among four input strings, at least two strings are not codewords and each has distance at least one. The total number of errors over the eight equiprobable pairs (x, j) is at least two. Thus the score is at most $6/8$. Averaging over the coins preserves the bound. Transmitting x_1 and guessing x_2 achieves $3/4$. □

Theorem 6.2 (Optimal one-qubit score)**.** *Under the above resource contract, the maximum quantum score is*

$$p_* = \frac{1 + 1/\sqrt{2}}{2} = \cos^2(\pi/8) > 3/4.$$

Proof. For achievability, encode

$$\rho_x = \frac{1}{2}\left(I + \frac{(-1)^{x_1} X + (-1)^{x_2} Z}{\sqrt{2}}\right).$$

The Bloch vector has unit norm. Measure X for $j = 1$ and Z for $j = 2$, identifying eigenvalue $(-1)^b$ with output b. Each input/query pair succeeds with probability p_*.

For optimality, any binary POVM defines an observable $A_j = E_{0|j} - E_{1|j} = \alpha_j I + \boldsymbol{a}_j \cdot \boldsymbol{\sigma}$ with $-I \le A_j \le I$, hence $\|\boldsymbol{a}_j\| \le 1$. Write $\rho_x = (I + \boldsymbol{r}_x \cdot \boldsymbol{\sigma})/2$ with $\|\boldsymbol{r}_x\| \le 1$. Uniform averaging cancels the α_j terms and gives

$$\begin{aligned} s &= \frac{1}{2} + \frac{1}{16} \sum_x \boldsymbol{r}_x \cdot ((-1)^{x_1} \boldsymbol{a}_1 + (-1)^{x_2} \boldsymbol{a}_2) \\ &\le \frac{1}{2} + \frac{\|\boldsymbol{a}_1 + \boldsymbol{a}_2\| + \|\boldsymbol{a}_1 - \boldsymbol{a}_2\|}{8} \le \frac{1}{2} + \frac{\sqrt{2}}{4}. \end{aligned}$$

The final inequality is Cauchy–Schwarz followed by the parallelogram identity: $(\|a+b\| + \|a-b\|)^2 \le 4(\|a\|^2 + \|b\|^2) \le 8$. Shared randomness only mixes strategies and cannot improve the bound. □

This separation is a standard random access code result [1]. It does not assert that one qubit permits recovery of both bits in one trial.

6.2 Noise and an exact distance to the classical class

Apply depolarization $\mathcal{N}_\eta(\rho) = \eta\rho + (1-\eta)I/2$, $0 \le \eta \le 1$, to the displayed quantum encoding. Let P_η denote the resulting conditional behavior. Each input/query pair has success probability

$$p_\eta = \frac{1 + \eta/\sqrt{2}}{2}.$$

This isotropic noise parameter is distinct from the earlier basis-specific dephasing parameter λ.

Theorem 6.3 (Exact operational distance)**.** *Define* $d(P, \mathcal{C}_1) = \inf_{Q \in \mathcal{C}_1} \max_{x,j} \|P(\cdot \mid x, j) - Q(\cdot \mid x, j)\|_{\mathrm{TV}}$. *Then*

$$\boxed{d(P_\eta, \mathcal{C}_1) = \max\left\{0, \frac{\eta}{2\sqrt{2}} - \frac{1}{4}\right\}.}$$

Proof. For any $Q \in \mathcal{C}_1$, the difference in probabilities of the success event at each input is bounded by conditional total variation. Averaging and using Theorem 6.1 yields

$$\max_{x,j} \|P_\eta - Q\|_{\mathrm{TV}} \ge s(P_\eta) - s(Q) \ge p_\eta - 3/4.$$

For the matching construction, use shared randomness to choose $k \in \{1, 2\}$ uniformly, send x_k, answer exactly when $j = k$, and otherwise output a fair bit. This classical strategy has success probability $3/4$ for *every* (x, j). For binary outputs, its conditional total-variation distance to P_η is $|p_\eta - 3/4|$. This attains the lower bound when $p_\eta \ge 3/4$. If $1/2 \le p_\eta \le 3/4$, mix this strategy with input-independent fair guessing, with weight $q = 4(p_\eta - 1/2) \in [0, 1]$. The mixture matches P_η exactly at every input. Thus the distance is zero in this case. □

Corollary 6.4. *This behavior violates the classical one-bit bound exactly when* $\eta > 1/\sqrt{2}$. *In the ideal case,*

$$d(P_1, \mathcal{C}_1) = \frac{\sqrt{2} - 1}{4} \approx 0.103553.$$

The exact classical simulation below threshold concerns the tested input/output behavior, not equality of quantum and classical internal states for all conceivable experiments.

6.3 Compatibility with a universal classical seed

Theorem 5.1 and Theorem 6.3 concern different feasible sets. A global table decoder can retain the full preparation input and answer later queries using it. Such access violates the one-bit contract. With two transmitted classical bits, the decoder can retain x and reproduce P_η by flipping the requested bit with probability $1 - p_\eta$. Thus the separation certifies a restricted resource advantage, not impossibility of unrestricted classical simulation. Seed entropy, persistent memory size, and communication capacity must not be identified without proof.

7 A falsifiable experimental protocol

Use fresh independent uniform x, j on every trial. Enforce the preparation/query boundary and audit every input-dependent carrier, including device memory, timing, classical labels, and postselection. Record every trial, including losses under a predefined scoring rule. Fix the sample size and analysis before collecting data.

Let Z_t be the success indicator. If the classical resource contract holds conditionally on every past history, then $\mathbb{E}[Z_t \mid \mathcal{F}_{t-1}] \leq 3/4$. The exponential bounded-difference argument gives

$$\Pr\left\{\frac{1}{N}\sum_{t=1}^{N} Z_t \geq 3/4 + \epsilon\right\} \leq e^{-2N\epsilon^2}.$$

Indeed, conditional Hoeffding's lemma bounds the moment-generating function of each centered Z_t by $e^{u^2/8}$; iteration and Markov's inequality give $e^{-uN\epsilon + Nu^2/8}$, minimized at $u = 4\epsilon$. For a fixed observed excess ϵ, $N \geq \ln(1/\alpha)/(2\epsilon^2)$ is sufficient for this null-tail bound to be at most α. This is a significance bound, not a guarantee of statistical power. Optional stopping requires a separately valid sequential procedure.

An observed excess can reflect either violation of the classical bound under the contract or failure of the contract itself. A human remembering two presented bits is not constrained to a one-bit internal memory merely because the final answer is one bit. Consequently, ordinary cognitive random-access performance alone does not establish quantum neural dynamics.

8 What remains open

The benchmark closes a precise version of the resource-separation question. It does not close the consciousness hypothesis. A further research program must identify a physically defensible bottleneck, calibrate operations independently, compare classical models with adequate memory, and test out-of-sample predictions. For the two-qubit architecture one must derive a comparison appropriate to its total available resources rather than borrowing the one-qubit bound without restriction.

A potentially substantive new result would be a justified separation for a new causal architecture, a sharp entropy/communication tradeoff for simultaneous adaptive policies, or an independently validated physiological prediction. None is asserted here. The relation between the accessibility functional and subjective experience remains an empirical and conceptual question.

Verification and provenance

All displayed propositions and benchmark bounds have explicit mathematical proofs in this note. A local exhaustive enumeration checked all $16 \times 16 = 256$ deterministic one-bit encoder/decoder pairs, giving maximum score 6/8. The enumeration corroborates the finite

classical case; the proof covers mixtures as well. No new biological measurements were performed. No claim of a Lean build or an axiom audit is made for this manuscript. It was drafted with AI assistance for author review; references and novelty claims require normal scholarly scrutiny before submission.

References

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